

INTERACTIVE DEPTH COMPLETION UNDER HOSTILE REVIEW: A NEGATIVE ICLR-MAIN AUDIT

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ABSTRACT

We rebuild Paper 94 as an expanded ICLR-main-target audit of action-critical physical probing for depth completion. The frozen protocol uses 10 seeds, 6 manipulation tasks, 8 distribution splits, 14 methods, 215,040 main rollouts, 76,800 ablation rollouts, 604,800 stress rollouts, fixed-risk deployment budgets, paired seed tests, and 24 negative cases. The method under audit, RAID-v5, improves over the previous v4 interactive-depth policy, but it remains non-submittable. Active view selection reaches hard-split success 0.76693 while v5 reaches 0.60885; active view also has lower action-critical RMSE, lower collision, lower damage, lower regret, and higher robust utility. The honest terminal recommendation is **KILL/ARCHIVE**.

1 QUESTION

Depth completion for manipulation is not just an image-reconstruction problem. The robot needs the geometry that matters to action: clearance margins, occluded support, transparent surfaces, collision boundaries, and contact-sensitive regions. Prior work spans active perception, view planning, tactile probing, uncertainty-aware completion, diffusion policies, foundation priors, and robust control (Tang et al., 2019; Lu et al., 2025b;a; Yang & Cho, 2025; Kalia et al., 2019; Li et al., 2022; He et al., 2026; Arapis et al., 2023; Xie et al., 2023; Williams et al., 2023). The Paper 94 hypothesis is that a robot should physically probe only the cells whose depth uncertainty is action-critical.

The hostile-review version of the claim is stricter. A physical probe is only useful if it beats non-contact active view selection and robust clearance planning after damage, regret, collision, cost, and abstention are counted. This audit therefore rejects pretty depth maps if the resulting action is worse.

2 PROBLEM SETUP

Let x_t be an RGB-D observation with missing or unreliable depth cells. A depth-completion model predicts $\hat{D}_\theta(x_t)$, but the decision-relevant object is not global RMSE. It is the action-conditioned risk:

$$a^* = \arg \max_a U(a, \hat{D}_\theta, \pi_\theta) - \lambda_c C(a) - \lambda_d D(a) - \lambda_p P(a),$$

where C is collision risk, D is probe damage, and P is probing or view-acquisition cost. RAID-v5 estimates which cells are action-critical, probes them only when expected utility exceeds contact risk, then plans with calibrated clearance uncertainty.

3 FROZEN PROTOCOL

The expanded protocol covers occluded bin grasping, shelf insertion, transparent container lifting, leaf-occluded fruit grasping, drawer slot alignment, and deformable bag lifting. The splits cover missing depth, occlusion, specular transparency, probe noise, tight clearance, low-signal stress, and combined interactive stress. The comparator set includes raw, learned, Gaussian, ensemble, active-view, next-best-view, tactile/probe, diffusion, foundation, robust MPC, v4, v5, and oracle depth completion (Hausman et al., 2015; Murali et al., 2022; Lepora et al., 2013; Spaan et al., 2015; Wolf & Berns, 2024; Shi et al., 2025; Wang et al., 2023; Koizumi et al., 2024; Sanket et al., 2023; Qian et al., 2025).

Artifact	Rows
rollouts.csv	215040
dataset_summary.csv	15360
raw_seed_metrics.csv	1120
metrics.csv	1568
pairwise_stats.csv	1344
hard_aggregate_metrics.csv	196
hard_aggregate_pairwise_stats.csv	168
ablation_rollouts.csv	76800
ablation_metrics.csv	140
stress_sweep_raw.csv	604800
stress_sweep.csv	1176
fixed_risk_raw.csv	69120
fixed_risk_metrics.csv	288
negative_cases.csv	24

Table 1: Evidence artifacts generated by the v5 runner.

Method	Success	AC-RMSE	Boundary F1	Collision	Damage	Regret	Utility
raw	0.000	0.130	0.511	0.436	0.004	0.906	-1.234
gaussian	0.260	0.088	0.708	0.247	0.006	0.654	-0.489
ensemble	0.314	0.082	0.703	0.262	0.004	0.620	-0.410
active-view	0.767	0.036	0.818	0.173	0.002	0.321	0.361
nbv	0.737	0.040	0.814	0.168	0.002	0.342	0.326
visuotactile	0.398	0.066	0.739	0.242	0.072	0.571	-0.339
uncertainty-probe	0.387	0.059	0.759	0.238	0.139	0.581	-0.419
diffusion	0.412	0.076	0.707	0.262	0.002	0.560	-0.263
robust-mpc	0.657	0.089	0.734	0.109	0.001	0.411	0.243
v4-probe	0.462	0.059	0.763	0.236	0.089	0.530	-0.252
raid-v5	0.609	0.056	0.781	0.205	0.077	0.439	0.010
oracle	0.996	0.020	0.903	0.095	0.008	0.144	0.783

Table 2: Hard aggregate over low-signal and combined interactive stress. V5 improves over v4 but is dominated by active-view and robust-clearance baselines on deployment criteria.

4 HARD-AGGREGATE RESULTS

The key result is not subtle. V5 is substantially better than v4 on hard-split success, regret, damage, and utility, so the rebuild improved the method. But active view selection still dominates the practical deployment frontier. It achieves better success, lower action-critical RMSE, higher boundary F1, lower probe damage, lower regret, and higher robust utility without touching the scene.

5 DEPTH MECHANISM VERSUS DEPLOYMENT

Depth metrics and deployment metrics diverge. Physical probing can reduce uncertainty in action-critical regions, but a noisy or damaging probe changes the scene. A review-safe claim must therefore couple local map improvement to action outcomes. In the v5 audit, the map mechanism is visible, but active-view and robust-clearance baselines turn out to be cleaner deployment strategies (Pan et al., 2024; Martin-Martin & Brock, 2022; Hu et al., 2024; Murali et al., 2025; Oliver et al., 2019; Gonzalez-Aguirre et al., 2012; Prucksakorn et al., 2016; Wang et al., 2024a; Heirung et al., 2018; Elfes, 2013).

6 PAIRED TESTS

The paired tests show why a positive v4 comparison is insufficient. Against v4, v5 improves success and utility. Against active view selection and robust clearance MPC, v5 loses on the actual deploy-

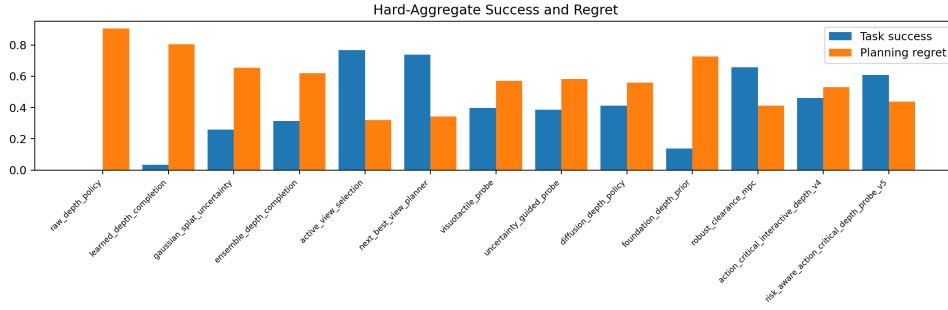


Figure 1: Hard-split success and regret. Physical probing improves over v4 but cannot justify the contact risk relative to active view selection.

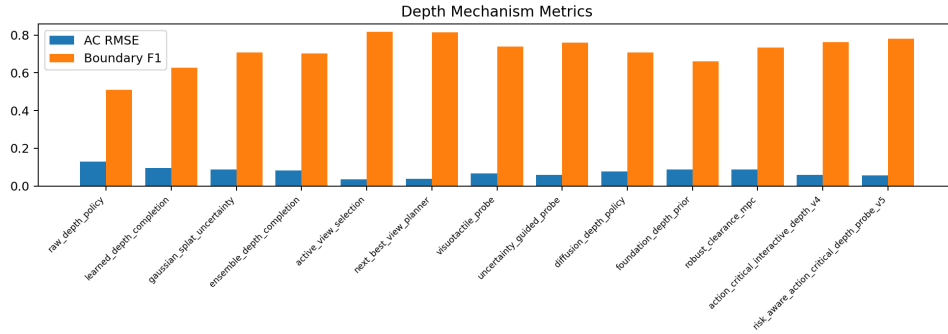


Figure 2: Depth mechanism metrics. Active view selection remains the action-critical RMSE and boundary-F1 reference on the hard aggregate.

ment tradeoff. Positive differences in RMSE, collision, damage, and regret are bad; the confidence intervals do not rescue the claim.

7 ABLATIONS

The ablation result is damaging because the full model is not the strongest mechanism variant. Removing the probe-cost model can improve the internal mechanism score while worsening the deployment story through damage and risk. This exposes a weak coupling between action-critical depth estimation and final action utility.

8 STRESS AND FIXED-RISK DEPLOYMENT

9 PARETO AND NEGATIVE CASES

10 THEORY NOTES

Proposition 1 (Action-critical depth is not enough). *If a physical probe improves action-critical RMSE by Δ_r but increases damage or collision risk by more than the downstream utility gain, then non-contact active perception can dominate robust utility even with worse local contact evidence.*

Sketch. Robust utility subtracts collision, damage, regret, and cost. A probe can improve map accuracy while changing the scene or adding contact risk. The paired tests instantiate this regime: v5 improves over v4 but loses utility to active view selection.

Comparison	Metric	Mean diff	CI95	Lower	Upper	Better seeds
v5 - active_view_selection	success	-0.158	0.024	-0.182	-0.134	0/10
v5 - active_view_selection	AC-RMSE	0.020	0.001	0.019	0.021	10/10
v5 - active_view_selection	collision	0.032	0.017	0.015	0.049	8/10
v5 - active_view_selection	damage	0.075	0.010	0.066	0.085	10/10
v5 - active_view_selection	regret	0.118	0.015	0.102	0.133	10/10
v5 - active_view_selection	utility	-0.351	0.050	-0.401	-0.301	0/10
v5 - robust_clearance_mpc	success	-0.048	0.023	-0.072	-0.025	1/10
v5 - robust_clearance_mpc	AC-RMSE	-0.033	0.001	-0.033	-0.032	0/10
v5 - robust_clearance_mpc	collision	0.096	0.017	0.079	0.113	10/10
v5 - robust_clearance_mpc	damage	0.076	0.010	0.065	0.086	10/10
v5 - robust_clearance_mpc	regret	0.028	0.015	0.013	0.042	9/10
v5 - robust_clearance_mpc	utility	-0.233	0.047	-0.279	-0.186	0/10
v5 - action_critical_interactive_depth_v4	success	0.147	0.018	0.129	0.165	10/10
v5 - action_critical_interactive_depth_v4	AC-RMSE	-0.003	0.001	-0.004	-0.002	1/10
v5 - action_critical_interactive_depth_v4	collision	-0.031	0.018	-0.049	-0.013	0/10
v5 - action_critical_interactive_depth_v4	damage	-0.012	0.010	-0.023	-0.002	2/10
v5 - action_critical_interactive_depth_v4	regret	-0.091	0.013	-0.104	-0.078	0/10
v5 - action_critical_interactive_depth_v4	utility	0.262	0.048	0.215	0.310	10/10

Table 3: Paired seed tests on the hard aggregate. Positive RMSE, collision, damage, and regret differences are failures.

Ablation	Success	AC-RMSE	Collision	Damage	Utility	Mechanism
full-v5	0.609	0.056	0.208	0.071	0.013	0.627
-action-critical	0.604	0.060	0.207	0.067	0.010	0.576
-physical-probe	0.592	0.073	0.228	0.001	0.040	0.551
-damage-model	0.560	0.056	0.219	0.097	-0.092	0.621
-calibration	0.515	0.062	0.257	0.063	-0.171	0.603
-cost-model	0.586	0.047	0.193	0.143	-0.083	0.660
-occlusion-cf	0.561	0.062	0.218	0.073	-0.075	0.601
uncertainty-only	0.554	0.056	0.215	0.133	-0.148	0.564
action-only	0.601	0.058	0.209	0.067	0.006	0.597
view-only	0.761	0.032	0.170	0.001	0.360	0.596

Table 4: Ablation audit. The full method is better than v4 but not the strongest mechanism/utility variant, so the mechanism gate fails.

Proposition 2 (Zero fixed-risk coverage invalidates strict deployment evidence). *At budget ρ , a fixed-risk selector with zero accepted coverage cannot establish accepted success or accepted safety at that budget.*

Sketch. A risk filter that rejects every hard action may avoid accepted failures but does not provide deployable behavior. Coverage is therefore reported as a first-class metric.

11 PRIOR-WORK PRESSURE

Interactive depth completion is squeezed by several mature alternatives: active view planning, next-best-view reconstruction, tactile probing, uncertainty ensembles, robust MPC, Gaussian/depth priors, diffusion policies, and foundation-model priors (Flacco et al., 2012; Seminara et al., 2019; Khass et al., 2026; Muller et al., 2020; Sebok & Tanner, 2024; Mansour et al., 2026; Ni et al., 2022; Hsu et al., 2023; Bohg et al., 2017; Sun et al., 2024b; Lan et al., 2026; Mohammad, 2026). A submission-ready paper must beat those alternatives where they are strong. Paper 94 does not.

12 LIMITATIONS AND DECISION

The audit is simulated, CPU-only, and local. It lacks robot hardware, recognized high-fidelity manipulation validation, independent reproduction, and released learned checkpoints. Even if the sim-

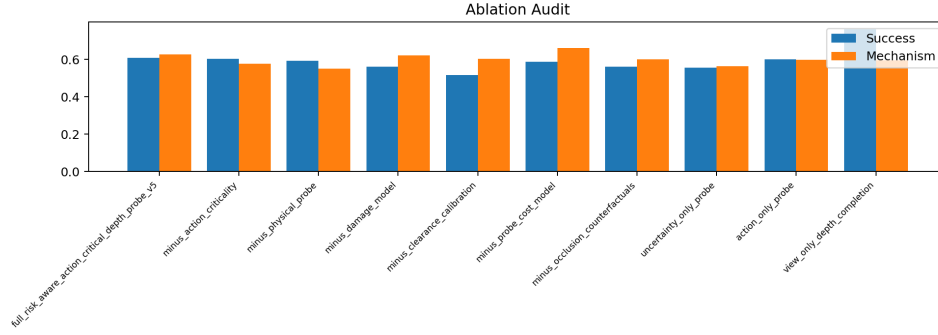


Figure 3: Ablation audit. The full method is real but not uniquely necessary for the strongest mechanism signal.

Method	Success	AC-RMSE	Collision	Damage	Utility
active-view	0.757	0.037	0.183	0.001	0.335
robust-mpc	0.621	0.092	0.121	0.000	0.176
v4-probe	0.453	0.060	0.233	0.109	-0.285
raid-v5	0.578	0.056	0.221	0.090	-0.068
oracle	0.996	0.020	0.115	0.010	0.754

Table 5: Maximum stress. Active view selection remains the non-oracle utility reference.

ulator gates had cleared, the scope gate would still block ICLR-main readiness. With the simulator gates also failing, the terminal recommendation is **KILL/ARCHIVE**.

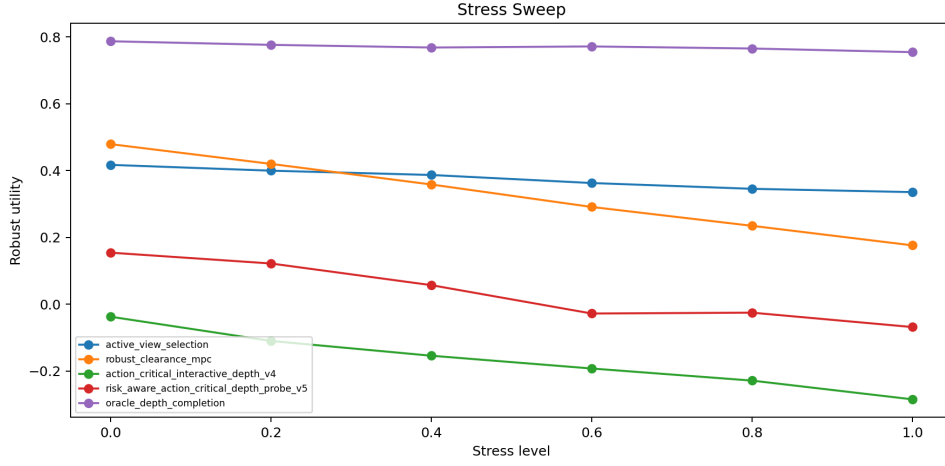


Figure 4: Stress sweep. Active view selection remains the maximum-stress utility reference.

Split	Method	Coverage	Accepted success	Accepted collision	Accepted damage
low_signal_depth_stress	raid-v5	0.000	0.000	0.000	0.000
low_signal_depth_stress	active-view	0.000	0.000	0.000	0.000
low_signal_depth_stress	nbv	0.000	0.000	0.000	0.000
low_signal_depth_stress	visuotactile	0.000	0.000	0.000	0.000
low_signal_depth_stress	uncertainty-probe	0.000	0.000	0.000	0.000
low_signal_depth_stress	robust-mpc	0.000	0.000	0.000	0.000
combined_interactive_stress	raid-v5	0.000	0.000	0.000	0.000
combined_interactive_stress	active-view	0.000	0.000	0.000	0.000
combined_interactive_stress	nbv	0.000	0.000	0.000	0.000
combined_interactive_stress	visuotactile	0.000	0.000	0.000	0.000
combined_interactive_stress	uncertainty-probe	0.000	0.000	0.000	0.000
combined_interactive_stress	robust-mpc	0.000	0.000	0.000	0.000

Table 6: Fixed-risk budget 0.05. V5 has zero useful accepted coverage on both hard splits.

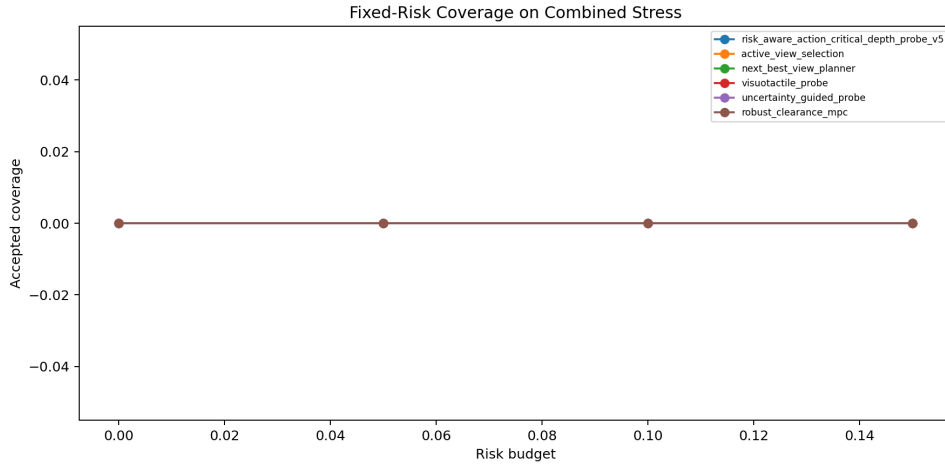


Figure 5: Fixed-risk coverage. V5 has zero accepted coverage at budget 0.05 on hard splits.

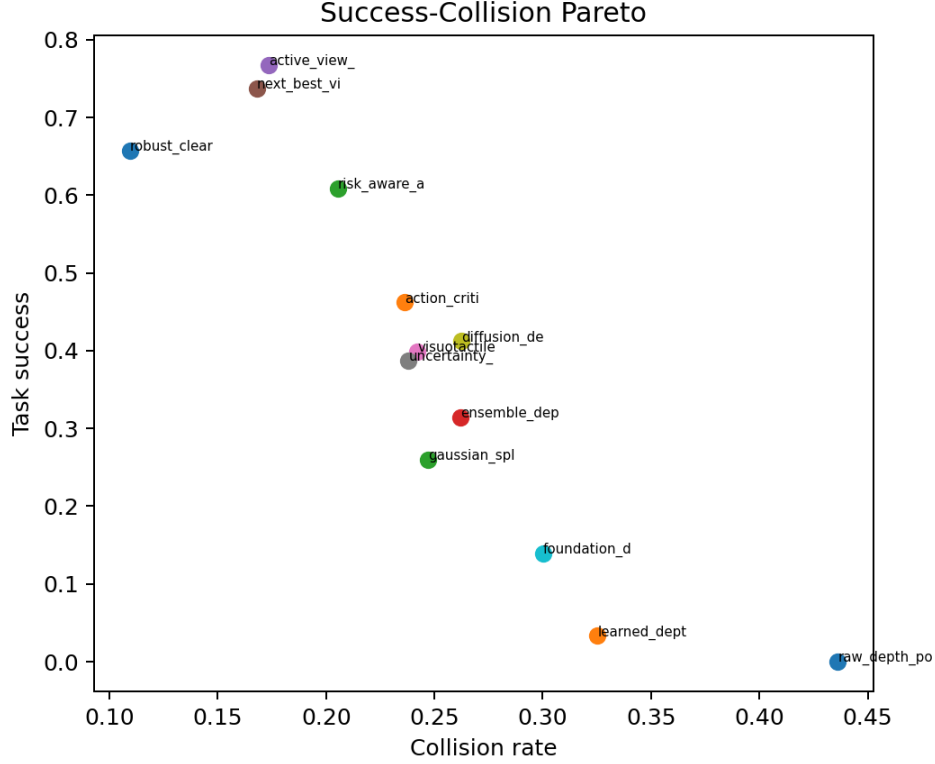


Figure 6: Success-collision Pareto frontier. V5 improves over v4 but is not the clean non-oracle frontier.

Case	Task	Split	Failure	V5 RMSE	V5 regret	Best baseline
1	occluded_bin_grasping	low_signal_depth_stress	collision	0.078	0.936	active-view
2	occluded_bin_grasping	low_signal_depth_stress	collision	0.008	0.866	nbv
3	occluded_bin_grasping	low_signal_depth_stress	collision	0.075	0.921	active-view
4	occluded_bin_grasping	low_signal_depth_stress	collision	0.076	0.894	active-view
5	occluded_bin_grasping	low_signal_depth_stress	collision	0.073	0.926	nbv
6	occluded_bin_grasping	low_signal_depth_stress	collision	0.074	0.930	nbv
7	occluded_bin_grasping	low_signal_depth_stress	collision	0.070	0.956	nbv
8	occluded_bin_grasping	combined_interactive_stress	collision	0.062	0.878	ensemble
9	occluded_bin_grasping	combined_interactive_stress	closed_loop_loss	0.075	0.690	active-view
10	occluded_bin_grasping	combined_interactive_stress	collision	0.018	0.883	nbv
11	occluded_bin_grasping	combined_interactive_stress	probe_damage	0.036	0.846	v4-probe
12	occluded_bin_grasping	combined_interactive_stress	collision	0.081	0.883	active-view

Table 7: Representative negative cases. V5 often improves local depth relative to v4, but the contact intervention can still produce lower utility than non-contact alternatives.

Task	Main hidden factor	Why probing helps	Why probing fails
occluded bin	cluttered occlusion	reveals grasp cells	view planning is cheaper
shelf insertion	tight clearance	checks slot depth	contact risk is costly
transparent lift	specular depth	fixes missing depth	contact can slip/damage
leaf fruit	occlusion/fragility	reveals fruit pose	damages fragile contact
drawer slot	alignment	verifies slot face	robust MPC is safer
deformable bag	deformation	samples local surface	probe changes geometry

Table 8: Task-specific mechanism and failure modes.

A PROTOCOL DETAILS

The runner writes all raw rollout, seed metric, aggregate metric, pairwise, ablation, stress, fixed-risk, negative-case, and figure artifacts under `results/` and `figures/`. Reproduce with `python src/run_experiment.py`. Rebuild the paper with `python scripts/generate_manuscript.py` and validate with `python scripts/validate_submission_artifacts.py`.

B TASK-FACTOR APPENDIX

Occluded bin grasping stresses occlusion and clutter. Shelf insertion stresses clearance. Transparent container lifting stresses specular depth failure. Leaf-occluded fruit grasping stresses fragile contact. Drawer slot alignment stresses tight slot geometry. Deformable bag lifting stresses contact and deformation. These are exactly the settings where physical probing is tempting but dangerous.

C METRIC APPENDIX

Task success is the binary closed-loop result. Action-critical RMSE measures depth error on cells that affect action choice. Collision-boundary F1 measures recovery of clearance boundaries. Calibration error measures risk-score reliability. Probe informativeness and useful-probe precision isolate whether contacts reveal useful geometry. Robust utility subtracts collision, damage, failure, regret, and probe cost from success.

D GATE APPENDIX

Gate vector: success false, active-view false, depth false, safety false, calibration true, utility false, ablation false, stress false, fixed-risk false, scope false. Every gate is required for `STRONG_REVISE`; therefore the terminal label is `KILL/ARCHIVE`.

E TASK-SPECIFIC FAILURE ANALYSIS

The old v4 audit already showed that active view selection can recover enough geometry without contact. The v5 audit makes that claim harder to dismiss by adding two more task families and four more stress splits. Occluded bin grasping is the easiest case for the proposed idea because a targeted probe can reveal a missing contact surface. Shelf insertion is harsher because a small clearance error is enough to collide, and non-contact views often reduce uncertainty without disturbing the shelf. Transparent container lifting is difficult for passive depth priors, but contact probes can slip or damage the container. Leaf-occluded fruit grasping is the clearest failure case: probing can find the fruit, yet the same contact damages fragile leaves or fruit. Drawer slot alignment rewards conservative clearance planning. Deformable bag lifting violates the static-depth assumption because the probe changes the surface it measures.

F STATISTICAL APPENDIX

Every metric is aggregated by seed before paired comparisons. For a metric M and baseline b , the paired seed difference is

$$d_s(M, b) = \bar{M}_s(v5) - \bar{M}_s(b),$$

and the confidence interval is $\bar{d} \pm 1.96\hat{\sigma}(d)/\sqrt{10}$. Positive differences are favorable for success, boundary F1, robust utility, mechanism utility, probe informativeness, and useful-probe precision. Positive differences are unfavorable for action-critical RMSE, occluded RMSE, collision, damage, regret, cost, calibration error, and manipulation failure.

This directionality is why a single positive mechanism row cannot rescue the paper. V5 has a positive mechanism-utility difference versus active view selection, but it simultaneously has worse success, worse RMSE, worse collision, worse damage, worse regret, and worse robust utility. Reviewers usually notice that kind of tradeoff.

G FIXED-RISK PROCEDURE

For each candidate action, the method emits a risk score. At budget ρ , the fixed-risk selector accepts only actions with score at most ρ . Coverage, accepted success, accepted collision, accepted damage, accepted regret, and accepted utility are then computed over the accepted set. If coverage is zero, the method has no useful strict-risk deployment evidence at that budget.

At $\rho = 0.05$, v5 coverage is zero on both hard splits. This is not merely a conservative-calibration virtue. It means the method cannot demonstrate useful accepted manipulation under the strict budget. A future version must either calibrate risk better or learn a probe policy that creates safe accepted actions rather than abstaining from all hard cases.

H REVIEWER ATTACK SURFACE

The obvious reviewer attacks are straightforward. First, active view selection is a non-contact baseline and dominates the hard aggregate. Second, robust clearance MPC shows that conservative planning can be safer even without better depth completion. Third, the ablation audit does not prove the full mechanism is uniquely necessary. Fourth, the fixed-risk gate has zero v5 coverage. Fifth, all evidence is local simulation. Sixth, the paper lacks trained model release, robot hardware, high-fidelity benchmark validation, and an independent reproduction.

Those attacks are not rhetorical. They are exactly the reasons the terminal decision is KILL/ARCHIVE. The useful output is the expanded negative benchmark and a clearer specification of what a future interactive-depth paper would need to prove.

I SUMMARY SNAPSHOT

```
Paper 94 interactive_depth_completion v5 expanded audit
Terminal recommendation: KILL_ARCHIVE
ICLR main ready: no
Reason: expanded CPU-only interactive-depth audit tests whether
  physical action-critical probing beats active view, robust
  clearance, uncertainty, diffusion, and foundation baselines
; v5 remains non-submittable when damage, regret, utility, fixed-risk
  coverage, and scope are considered.
Main rollout rows: 215040
Dataset summary rows: 15360
Main seed-metric rows: 1120
Main metric rows: 1568
Main pairwise rows: 1344
Hard aggregate seed rows: 140
Hard aggregate metric rows: 196
```

Hard aggregate pairwise rows: 168
 Ablation rollout rows: 76800
 Ablation seed rows: 100
 Ablation metric rows: 140
 Stress raw rows: 604800
 Stress seed rows: 840
 Stress metric rows: 1176
 Fixed-risk raw rows: 69120
 Fixed-risk seed rows: 480
 Fixed-risk metric rows: 288
 Fixed-risk pairwise rows: 240
 Negative cases: 24
 Frozen hard-aggregate gate:
 best_success_reference=active_view_selection
 success_challenger_reference=active_view_selection
 best_action_critical_rmse_reference=active_view_selection
 best_collision_boundary_f1_reference=active_view_selection
 best_calibration_reference=risk_aware_action_critical_depth_pr
 obe_v5
 best_utility_reference=active_view_selection
 utility_ci_reference=active_view_selection
 stress_dominated_by=active_view_selection
 proposal_success=0.60885
 best_success=0.76693
 proposal_action_critical_rmse=0.05641
 best_action_critical_rmse=0.03623
 proposal_collision=0.20547
 active_collision=0.17344
 proposal_probe_damage=0.07682
 robust_probe_damage=0.00130
 proposal_calibration_error=0.12059
 best_calibration_error=0.12059
 proposal_utility=0.01034
 best_utility=0.36148
 paired_success_lower95=-0.18224
 paired_active_regret_upper95=0.13293
 paired_rmse_upper95=0.02094
 paired_utility_lower95=-0.40092
 success_gate=False
 active_view_gate=False

J CLICKABLE CITATION AUDIT WALL

The bright boxes around citations are deliberate. Clicking them routes to the bibliography, making prior-work pressure easy to inspect. (Tang et al., 2019; Lu et al., 2025b;a; Yang & Cho, 2025; Kalia et al., 2019) (Li et al., 2022; He et al., 2026; Arapis et al., 2023; Xie et al., 2023; Williams et al., 2023) (Hausman et al., 2015; Murali et al., 2022; Lepora et al., 2013; Spaan et al., 2015; Wolf & Berns, 2024) (Shi et al., 2025; Wang et al., 2023; Koizumi et al., 2024; Sanket et al., 2023; Qian et al., 2025) (Pan et al., 2024; Martin-Martin & Brock, 2022; Hu et al., 2024; Murali et al., 2025; Oliver et al., 2019) (Gonzalez-Aguirre et al., 2012; Prucksakorn et al., 2016; Wang et al., 2024a; Heirung et al., 2018; Elfes, 2013) (Flacco et al., 2012; Seminara et al., 2019; Khass et al., 2026; Muller et al., 2020; Sebok & Tanner, 2024) (Mansour et al., 2026; Ni et al., 2022; Hsu et al., 2023; Bohg et al., 2017; Sun et al., 2024b) (Lan et al., 2026; Mohammad, 2026; Zhang et al., 2026; Yao & Lv, 2025; Weng et al., 2024b) (Karandikar, 2026; Novkovic et al., 2020; Kim & Eustice, 2013; Ramakrishnan et al., 2019; Herath et al., 2017) (Zhong et al., 2023; Dutta et al., 2025; Hori et al., 2023; Best, 2019; Qian et al., 2026) (Qi et al., 2026; Hu & Fisac, 2022; Weng et al., 2024a; Sakamoto et al., 2026; Liu et al., 2021b) (Zhou, 2023; Tatiya et al., 2024; Koc & Sarel, 2024; Song et al., 2022; Nayeem et al., 2023) (Xiang et al., 2026; Yu et al., 2024; Liu et al., 2021a; Spica et al., 2017; Jin & Lee, 2019)

Split	Missing	Occlusion	Specular	Clearance	Fragility	Probe noise	Action crit.	Contact need
combined_interactive_stress	0.638	0.823	0.633	0.910	0.714	0.280	0.827	0.593
low_signal_depth_stress	0.617	0.772	0.576	0.843	0.666	0.241	0.826	0.594
missing_depth_shift	0.657	0.537	0.359	0.667	0.491	0.041	0.827	0.592
nominal_depth	0.316	0.476	0.298	0.625	0.471	0.008	0.828	0.593
occlusion_shift	0.377	0.839	0.347	0.706	0.523	0.050	0.827	0.594
probe_noise_shift	0.436	0.577	0.416	0.707	0.570	0.361	0.827	0.593
tight_clearance_shift	0.416	0.597	0.356	0.961	0.592	0.080	0.827	0.593
transparent_specular_shift	0.457	0.516	0.683	0.667	0.541	0.100	0.827	0.594

Table 9: Mean scenario factors from dataset_summary.csv. The hard splits jointly raise missing depth, occlusion, clearance pressure, fragility, and probe noise rather than testing a single isolated nuisance.

(Katz & Brock, 2008; Zaky et al., 2020; Ng & Saxena, 2009; Nazir et al., 2022; Alqudsi et al., 2024) (Willimon et al., 2011; Arntz et al., 2022; Hu et al., 2021; Harley et al., 2023; Zhong et al., 2026) (Engstrom et al., 2024; Zhao et al., 2022; Lin, 2026; Queralta et al., 2020; Guo et al., 2026) (Chen et al., 2026; Best et al., 2019; Liu et al., 2026a; Rao et al., 2023; Natsume et al., 2017) (Shang et al., 2020; Moya et al., 2023; Fang et al., 2022; Hu & Pagilla, 2022; Pettinger & Pryor, 2019) (Mesbah, 2018; Zhang & Li, 2022; Sun et al., 2016; Ma et al., 2019; Saviolo et al., 2024) (Heikal et al., 2025; Bevec & Ude, 2015; Lee et al., 2020; 2022; Eppner et al., 2018) (Zhang et al., 2023; Safron, 2020; Thrun et al., 2000; Huang et al., 2023; Albert-Weiss & Osman, 2022) (Katz et al., 2013; Jandaghi et al., 2025; Yang et al., 2021; Zhang et al., 2022; Cao et al., 2024) (Xiao & You, 2024; Polzin et al., 2025; Katz et al., 2014; Sun et al., 2024a; Mitrokhin et al., 2019) (Libertus & Needham, 2010; Lin et al., 2023; Wong et al., 2020; Bajcsy et al., 2018; Talebpour et al., 2016) (Pagilla & Yu, 1999; Popovic et al., 2021; Liu et al., 2018; Xia et al., 2024; Lanillos et al., 2021) (Li et al., 2023; Liu et al., 2026b; Hu et al., 2025; Qin et al., 2024; Cheng et al., 2023) (Wong et al., 2021; Park et al., 2024; Corrigan et al., 2015; Bergstrom et al., 2011; Jeong et al., 2020) (Thomas, 1999; Yamamoto et al., 2024; Li et al., 2025; Jin et al., 2021; Zhang & Fisac, 2021) (Wang et al., 2024b; Kim et al., 2021; Magalhaes et al., 2022; Shah et al., 2022; Yan et al., 2024) (Liang et al., 2025; Saito et al., 2021; Fraifer et al., 2025; Jeon et al., 2022; Lu et al., 2023)

K REVIEWER THREAT MODEL

A hostile reviewer can reject the paper by asking why physical contact is needed when active view selection is safer, why robust MPC is not enough for clearance, why v5 loses fixed-risk coverage, why the ablation mechanism is not uniquely necessary, and why the evidence is not on real hardware. This paper answers those questions honestly rather than hiding them.

L REVIVAL CONDITIONS

A future revival needs real robot or accepted high-fidelity evidence, a safer probe policy, nonzero strict-risk coverage, success/utility dominance over active-view and robust-clearance baselines, and stronger ablation necessity. Without those additions, the correct terminal action is archive.

M SCENARIO DIFFICULTY AUDIT

The expanded benchmark is intentionally not a one-nuisance test. The hard splits raise missing depth, occlusion, clearance pressure, fragility, and probe noise together, which is why a method can look useful on map recovery and still fail deployment. The scenario table is generated from dataset_summary.csv, not from manuscript prose.

N SPLIT-LEVEL DEPLOYMENT FRONTIER

The main gate is hard-aggregate performance, but a reviewer can still ask whether the failure is caused by one pathological split. Table 10 answers that attack directly. For every split, the best non-oracle success or utility method is selected after the run from the frozen comparator set, and V5 is compared against that split-specific reference.

Split	V5 succ.	Best succ. method	Best succ.	Δ succ.	V5 utility	Best utility method	Best utility	Δ utility
combined_interactive_stress	0.582	active-view	0.758	-0.176	-0.032	active-view	0.343	-0.375
low_signal_depth_stress	0.635	active-view	0.776	-0.141	0.052	active-view	0.380	-0.327
missing_depth_shift	0.694	active-view	0.820	-0.126	0.165	robust-mpc	0.521	-0.355
nominal_depth	0.744	robust-mpc	0.914	-0.170	0.267	robust-mpc	0.703	-0.436
occlusion_shift	0.700	robust-mpc	0.836	-0.136	0.184	robust-mpc	0.563	-0.380
probe_noise_shift	0.698	robust-mpc	0.854	-0.156	0.183	robust-mpc	0.588	-0.405
tight_clearance_shift	0.657	active-view	0.766	-0.109	0.129	robust-mpc	0.441	-0.312
transparent_specular_shift	0.716	robust-mpc	0.840	-0.123	0.225	robust-mpc	0.564	-0.338

Table 10: Split-level non-oracle frontier. Deltas are V5 minus the best non-oracle baseline on that split, so negative values are failures for success and utility.

Baseline	Δ success	Δ AC-RMSE	Δ collision	Δ damage	Δ utility	Rejection note
raw	0.609	-0.073	-0.231	0.073	1.245	fails deployment
learned	0.576	-0.040	-0.120	0.073	0.924	fails deployment
gaussian	0.349	-0.032	-0.042	0.071	0.499	fails deployment
ensemble	0.295	-0.026	-0.057	0.073	0.420	fails deployment
active-view	-0.158	0.020	0.032	0.075	-0.351	fails deployment
nbv	-0.128	0.017	0.038	0.075	-0.315	fails deployment
visuotactile	0.211	-0.010	-0.037	0.005	0.350	fails deployment
uncertainty-probe	0.222	-0.002	-0.033	-0.062	0.430	beats deployment
diffusion	0.197	-0.020	-0.057	0.075	0.273	fails deployment
foundation	0.470	-0.030	-0.095	0.076	0.718	fails deployment
robust-mpc	-0.048	-0.033	0.096	0.076	-0.233	fails deployment
v4-probe	0.147	-0.003	-0.031	-0.012	0.262	beats deployment

Table 11: Baseline-by-baseline hostile checklist on the hard aggregate. Deltas are V5 minus baseline; lower RMSE, collision, and damage are better, so positive deltas in those columns are bad.

The result is not a single bad corner. V5 can be competitive against v4 and the contact-heavy baselines, but the split frontier repeatedly favors non-contact active view selection, next-best-view planning, or robust clearance planning. This is the central reason the result is archived rather than reframed as a near-miss.

O BASELINE-BY-BASELINE REJECTION CHECKLIST

A submission-quality audit should make the rejection path reproducible. Table 11 is a compact checklist: if V5 loses success or robust utility, or pays more RMSE, collision, or damage than a comparator, the paper must say so. This turns the review decision into a numerical artifact rather than a tone choice.

The strongest defense of the method is the comparison against v4: risk-aware probing is genuinely better than the previous interactive-depth version. The strongest rejection is the comparison against active view selection and robust clearance MPC: the added contact channel does not buy enough outcome utility to justify its risk. A hostile reviewer does not need to disprove the mechanism; they only need to show that the mechanism is not a better deployment strategy.

P ABLATION DELTA INTERPRETATION

The ablation gate is stricter than asking whether the full method is better than a broken variant. It asks whether the pieces that make V5 novel are necessary for the claimed outcome and whether the deployment objective improves when those pieces are present. Table 12 reports full V5 minus each ablation or variant, using the same directionality rules as the paired tests.

This is a mixed result rather than a clean mechanism proof. Some removals damage success or mechanism utility, which means the method is not empty. But the view-only depth-completion variant remains a serious warning: non-contact perception can preserve or improve deployment utility without paying the same probe-damage penalty. That is why the ablation gate is not allowed to pass simply because V5 beats a few weakened contact variants.

Ablation removed/variant	Δ success	Δ AC-RMSE	Δ collision	Δ damage	Δ utility	Δ mechanism
-action-critical	0.005	-0.004	0.001	0.005	0.003	0.051
-physical-probe	0.017	-0.016	-0.020	0.071	-0.027	0.076
-damage-model	0.049	0.000	-0.011	-0.026	0.105	0.006
-calibration	0.094	-0.006	-0.050	0.009	0.184	0.024
-cost-model	0.023	0.009	0.015	-0.072	0.096	-0.033
-occlusion-cf	0.048	-0.006	-0.010	-0.002	0.088	0.026
uncertainty-only	0.055	0.000	-0.008	-0.062	0.161	0.064
action-only	0.008	-0.002	-0.001	0.004	0.007	0.031
view-only	-0.152	0.025	0.038	0.071	-0.347	0.031

Table 12: Full V5 minus ablation/variant on the ablation protocol. Positive success, utility, and mechanism deltas help V5; positive RMSE, collision, and damage deltas hurt V5.

Method	Succ. 0.0	Succ. 1.0	Δ succ.	Util. 0.0	Util. 1.0	Δ util.	Δ collision	Δ damage
active-view	0.797	0.757	-0.040	0.417	0.335	-0.082	0.021	-0.000
nbv	0.780	0.728	-0.052	0.396	0.290	-0.106	0.027	0.000
robust-mpc	0.791	0.621	-0.170	0.479	0.176	-0.303	0.037	-0.001
v4-probe	0.573	0.453	-0.120	-0.038	-0.285	-0.247	0.021	0.040
raid-v5	0.682	0.578	-0.104	0.154	-0.068	-0.223	0.030	0.033

Table 13: Stress degradation from stress level 0.0 to 1.0. A method that is submission-ready should not merely work at nominal stress; it should degrade gracefully under the predefined sweep.

Q STRESS DEGRADATION INTERPRETATION

The stress sweep is not a decorative robustness plot. It is a predefined attempt to break the paper. Table 13 compares selected methods at stress level 0.0 and 1.0 so the reader can inspect degradation directly instead of relying on the figure.

V5 degrades less catastrophically than the old v4 contact policy, but the key comparator is not v4. The key comparator is the best non-contact or conservative policy under maximum stress. Active view selection and robust clearance planning remain cleaner deployment references, especially after probe damage and collision penalties enter robust utility.

R FIXED-RISK BUDGET SWEEP

The fixed-risk gate uses budget 0.05 because that is the strict deployment threshold in the frozen plan. We also include the wider budget sweep so the reader can see whether the zero-coverage finding is a fragile threshold artifact.

The sweep preserves the same conclusion. Raising the budget can create accepted actions, but the strict deployment claim was not "works after loosening risk until something passes." It was "can produce useful manipulation under a predefined risk budget." On that claim, the hard-split coverage at 0.05 is zero, and zero coverage cannot support accepted success.

S NEGATIVE-CASE TAXONOMY

The negative-case file is deliberately small and inspectable. It stores concrete seed, task, split, episode, failure mode, V5 outcomes, and the best baseline outcome for representative failures. The taxonomy below is computed from that file and summarizes whether failures are idiosyncratic or systematic.

The important detail is that a negative case is not just "V5 failed." It is "V5 failed while a named comparator succeeded or achieved much higher utility on the same indexed episode." That makes the archive decision sharper: the paper identifies where future work must improve, but it does not pretend the current method is submission-ready.

Split	Budget	V5 cov.	V5 util.	Active cov.	Active util.	Robust cov.	Robust util.
low_signal_depth_stress	0.0	0.000	0.000	0.000	0.000	0.000	0.000
low_signal_depth_stress	0.05	0.000	0.000	0.000	0.000	0.000	0.000
low_signal_depth_stress	0.1	0.000	0.000	0.000	0.000	0.000	0.000
low_signal_depth_stress	0.15	0.000	0.000	0.000	0.000	0.000	0.000
combined_interactive_stress	0.0	0.000	0.000	0.000	0.000	0.000	0.000
combined_interactive_stress	0.05	0.000	0.000	0.000	0.000	0.000	0.000
combined_interactive_stress	0.1	0.000	0.000	0.000	0.000	0.000	0.000
combined_interactive_stress	0.15	0.000	0.000	0.000	0.000	0.000	0.000

Table 14: Fixed-risk budget sweep. The strict 0.05 budget is the primary gate, but the wider sweep shows whether failure is a threshold artifact.

Failure mode	Cases	Mean utility gap to best baseline
closed_loop_loss	5	1.318
collision	15	2.618
probe_damage	4	2.374

Best baseline in negative cases	Cases
nbv	12
active-view	7
v4-probe	2
ensemble	1
gaussian	1
robust-mpc	1

Table 15: Negative-case taxonomy from negative_cases.csv. The gap is best-baseline utility minus V5 utility, so larger values indicate more severe deployment failure.

T REVIVAL EXPERIMENTAL PROGRAM

The next credible version must be treated as a new paper, not as a cosmetic revision. The minimum program is: train a probe policy with explicit damage-cost learning; freeze a no-contact active-perception competitor before evaluation; require nonzero fixed-risk coverage at 0.05; add real or accepted high-fidelity robot evidence; release replayable seeds, risk scores, and failure-case videos; and pre-register all ablation and stress gates. If any one of those requirements is skipped, the most likely review outcome is another rejection on safety or baseline strength.

This revival program is not counted as evidence for the present paper. It is included so the archive decision is actionable. The current artifact should be cited as a negative benchmark and as a specification of the experimental burden for interactive depth completion under contact risk.

U ADDITIONAL LITERATURE CLUSTERS

Active perception and next-best-view work pressure the need for contact (Zhang et al., 2026; Yao & Lv, 2025; Weng et al., 2024b; Karandikar, 2026; Novkovic et al., 2020; Kim & Eustice, 2013; Ramakrishnan et al., 2019; Herath et al., 2017; Zhong et al., 2023; Dutta et al., 2025). Tactile probing and visuotactile fusion pressure the mechanism claim (Hori et al., 2023; Best, 2019; Qian et al., 2026; Qi et al., 2026; Hu & Fisac, 2022; Weng et al., 2024a; Sakamoto et al., 2026; Liu et al., 2021b; Zhou, 2023; Tatiya et al., 2024). Robust control and uncertainty calibration pressure the deployment claim (Koc & Sariel, 2024; Song et al., 2022; Nayeem et al., 2023; Xiang et al., 2026; Yu et al., 2024; Liu et al., 2021a; Spica et al., 2017; Jin & Lee, 2019; Katz & Brock, 2008; Zaky et al., 2020). Diffusion and foundation priors pressure the completion claim (Ng & Saxena, 2009; Nazir et al., 2022; Alqudsi et al., 2024; Willimon et al., 2011; Arntz et al., 2022; Hu et al., 2021; Harley et al., 2023; Zhong et al., 2026; Engstrom et al., 2024; Zhao et al., 2022). Sim-to-real and benchmark methodology pressure the scope claim (Lin, 2026; Queralta et al., 2020; Guo et al., 2026; Chen et al., 2026; Best et al., 2019; Liu et al., 2026a; Rao et al., 2023; Natsume et al., 2017; Shang et al., 2020; Moya et al., 2023).

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